32548-48730 Cybersecurity Lab 7

Configuration of Security Policy on Juniper SRX

Question 1. Provide explanation for your observation of the currently configured security policies?

Answer: The current security policies allow full access in both directions. From Internet to internal, the policy permits any source, any destination and any application. From Internal to Internet, the same any-any-any permit rule is applied. This means all traffic allowed without restriction, which is fine for lab testing but not secure for real networks.

```
[edit]
student@ROUTER_27# commit
commit complete

[edit]
student@ROUTER_27# show security policies
from-zone Internet to-zone Internet {
    policy All_Internet_Internet {
        match {
            source-address any;
            destination-address any;
            application any;
        }
        then {
            permit;
        }
    }
}
from-zone Internet to-zone Internal {
    policy All_Internet_Internal {
        match {
            source-address any;
            destination-address any;
            application any;
        }
        then {
            permit;
        }
    }
}
from-zone Internal to-zone Internet {
    policy All_Internal_Internet {
        match {
            source-address any;
            destination-address any;
            application any;
        }
        then {
            permit;
        }
    }
}

[edit]
student@ROUTER_27#
```

<u>Task 2: Configuring a Security Policy</u>

1. To add a security policy, enter edit security policies in configuration mode.
2. edit a security policy by assigning the “from-zone” as Internet and “to-zone” as Internal.

Example: edit from-zone Internet to-zone Internal

Note: The zones “Internet” and “Internal” are already defined and configured. You do not have to make new zones.

3. Generate a name for the policy by entering edit policy <name>
4. Generate match criteria including source address, destination address and application. Example: set match source-address <IP address/any>
   set match destination-address <IP address/any> set match application <application-name/any>

   For this task, use the application junos-ftp .
5. Assign an action for the security policy by entering set then <action> . Action can be permit, deny, reject, log, or count (or combination)
6. Enter up on the terminal.
7. Enter show to view security policies (Screenshot required).

```
[edit security policies from-zone Internet to-zone Internal policy Saif]
student@ROUTER_27# up

[edit security policies from-zone Internet to-zone Internal]
student@ROUTER_27# show
policy All_Internet_Internal {
    match {
        source-address any;
        destination-address any;
        application any;
    }
    then {
        permit;
    }
}
policy Saif {
    match {
        source-address any;
        destination-address any;
        application junos-ftp;
    }
    then {
        permit;
    }
}

[edit security policies from-zone Internet to-zone Internal]
student@ROUTER_27#
```

This security policy will only allow incoming FTP tra ic (from zone Internet to zone Internal). A security policy should also be created for outgoing FTP tra ic (From zone Internal to zone Internet)

8. Enter top on the terminal.
9. Enter edit security policies

10. Repeat Step 2, this time the “from-zone” should be “Internal” and “to-zone” should be “Internet”.
11. Create a new security policy along with match criteria and action used for the security policy you created for Incoming FTP trafic.
12. Take a screenshot of the Security Policy.

```
[edit security policies from-zone Internal to-zone Internet]
student@ROUTER_27# show
policy All_Internal_Internet {
    match {
        source-address any;
        destination-address any;
        application any;
    }
    then {
        permit;
    }
}
policy Saif {
    match {
        source-address any;
        destination-address any;
        application junos-ftp;
    }
    then {
        permit;
    }
}

[edit security policies from-zone Internal to-zone Internet]
student@ROUTER_27#
```

13. Enter top to navigate back to the top of the configuration hierarchy.

Question 2: Configure security policies between the zone “Internet” and “Internal”. Set the conditions to allow FTP access to the server?

Answer: In this configuration, a policy named Saif has been created under the security policies. The match conditions are set to any source address and any destination address, with the applications defined as junos-ftp to restrict the traffic to FTP only. Finally, the action in the then statement must be permit. This ensures that only FTP traffic is allowed from the Internet into the internal zone, while other applications are not covered by this policy.

Question 3 (Challenge): Configure security policies between zones “Internet” and “Internal”. Set the condition to reject telnet access to the Server. (Use the application “junos-telnet” for this task). Provide screenshots to support your answer.

Answer: The only change from the previous FTP policy is in the application field is set to junos-telnet, and then statement set to reject. This ensures telnet traffic is denied while FTP traffic remines unaffected

```
[edit security policies from-zone Internal to-zone Internet]
student@ROUTER_27# show
policy All_Internal_Internet {
    match {
        source-address any;
        destination-address any;
        application any;
    }
    then {
        permit;
    }
}
policy Saif {
    match {
        source-address any;
        destination-address any;
        application [ junos-ftp junos-telnet ];
    }
    then {
        reject;
    }
}

[edit security policies from-zone Internal to-zone Internet]
student@ROUTER_27#
```

```
student@ROUTER_27# show
policy All_Internet_Internal {
    match {
        source-address any;
        destination-address any;
        application any;
    }
    then {
        permit;
    }
}
policy Saif {
    match {
        source-address any;
        destination-address any;
        application [ junos-ftp junos-telnet ];
    }
    then {
        reject;
    }
}

[edit security policies from-zone Internet to-zone Internal]
student@ROUTER_27#
```